

PV239 – 06 API

Goals

- Web API concepts
- Client-server communication
- Generating API client and its usage to communicate with API

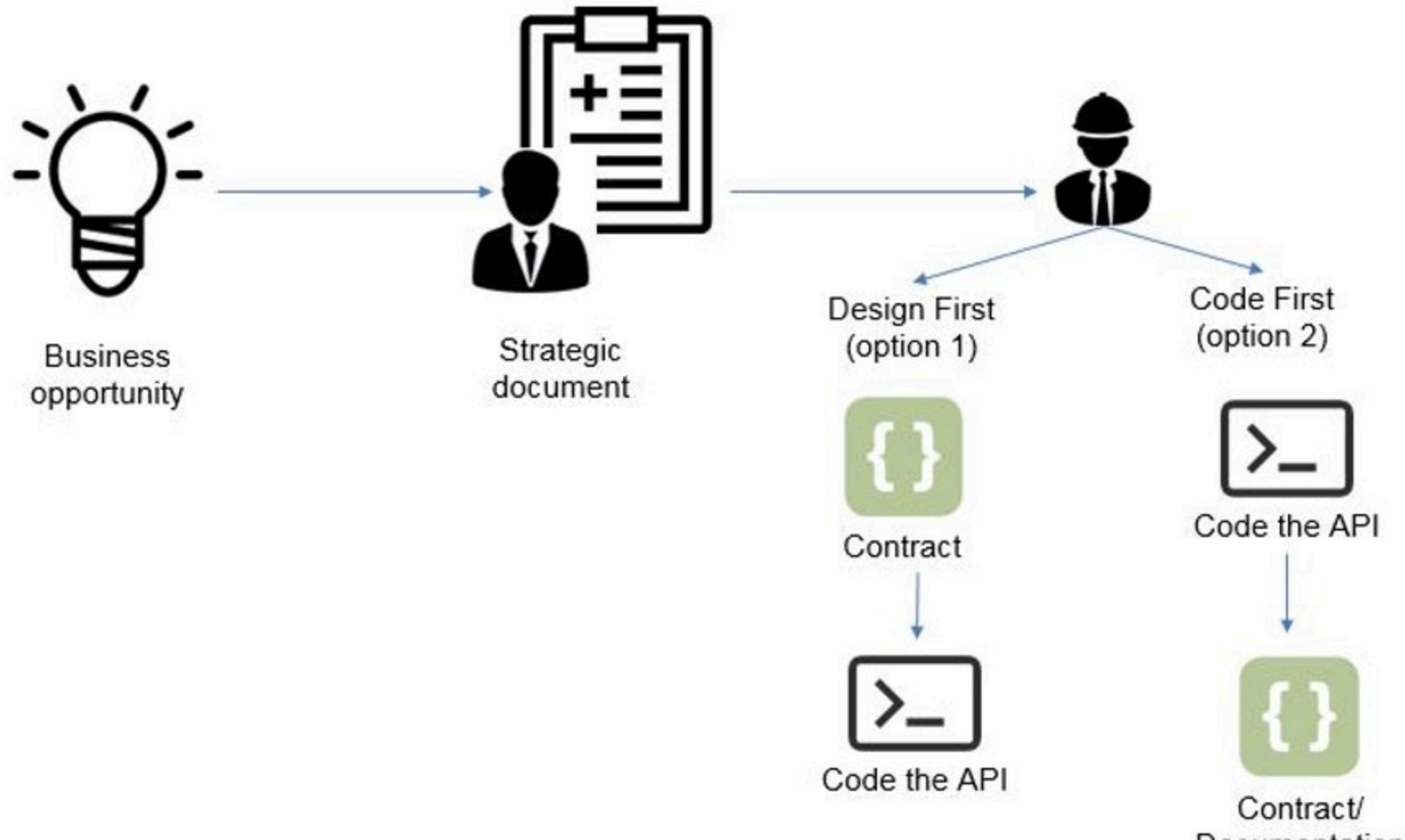
OpenAPI

- Open specification
- Standard for describing REST API
- Language and framework independent
- YAML/JSON
- OpenAPI map: <https://openapi-map.apihandyman.io/>

Swagger

- Tools for creating and working with OpenAPI specification
- Swagger Editor
 - YAML/JSON
- Swagger CodeGen
 - Generating client/server part
- Swagger UI
 - Interactive documentation

Approaches for creating API specification



Design-first approach

- SwaggerHub
 - <https://app.swaggerhub.com/>
- Portal for design first approach
- Supports hosting on own server

Code-first approach

- ASP .NET Core
- Library NSwag.AspNetCore

OPEN API Code-first

DEMO

Setup

- Nuget package NSwag.AspNetCore
- Add configuration:

```
services.AddOpenApiDocument(document =>
{
    document.Title = "CookBook API";
    document.DocumentName = "cookbook-api";
});
```

Setup

- Add middleware:

```
app.UseOpenApi();
```

- Add middleware for Swagger UI:

```
app.UseSwaggerUi3(settings =>
{
    settings.SwaggerRoutes.Add(
        new SwaggerUi3Route("CookBook API", "/swagger/cookbook-api/swagger.json"));
});
```

Swagger editor

- Online version <https://editor.swagger.io>
- Support for running locally
 - Docker image

Generating - NSwagStudio

- NSwag - CLI tool
- NSwagStudio - Desktop app
- <https://github.com/RSuter/NSwag>
- Settings:
 - Namespace
 - Set Operation Generation Mode
 - Path to output file

NSwagStudio

DEMO

Adding client to mobile app

- Add client registration to DI container:

```
serviceCollection.AddHttpClient<IIngredientsClient, IngredientsClient>(client =>
{
    client.BaseAddress = apiBaseUri;
});
For usage in emulators
URL for „localhost“: 10.0.2.2
```

Using generated client

DEMO

Processors

- Used for modifying the OpenAPI specification
- Can be used on multiple levels:
 - `IOperationProcessor` – modifies individual operations
 - `ISchemaProcessor` – modifies the schema
 - `IDocumentProcessor` – modifies the entire document

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